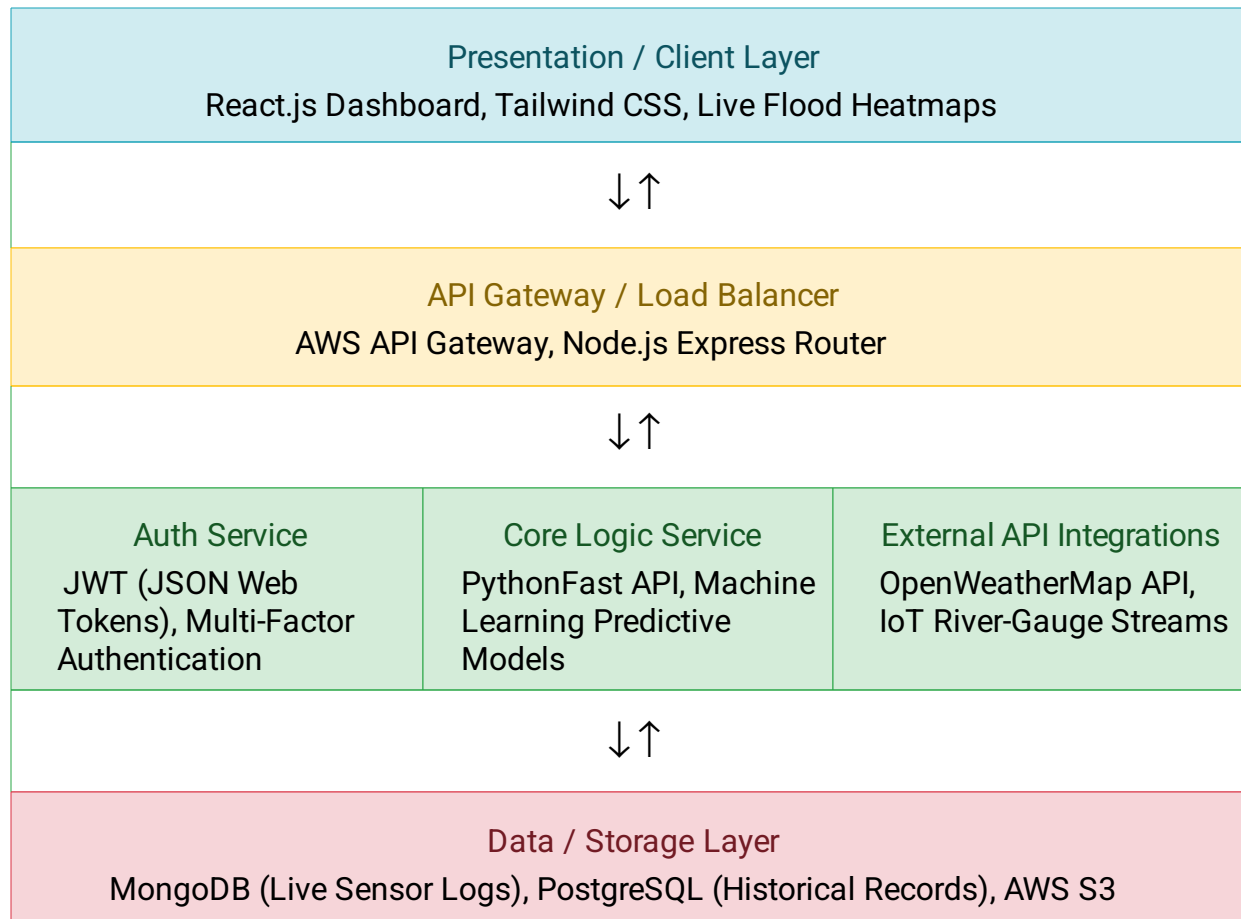


Solution Architecture

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Manages the user interface, rendering interactive dashboards, real-time flood heatmaps, and historical data logs directly to the disaster management officers.	React.js, Tailwind CSS
API Gateway	Acts as a single entry point that secures, routes, and load-balances incoming data traffic and client requests to the correct backend services.	AWS API Gateway, Express Server
Microservices	Auth Service: Validates secure officer logins using MFA. Core Logic Service: Runs Python-based predictive ML models to evaluate rainfall anomalies. External API Integrations: Syncs data from external weather stations.	Node.js, Python (FastAPI), JWT
Database	Handles data persistence, storing high-frequency incoming live sensor data streams and managing structured historical regional flood records.	MongoDB, PostgreSQL, AWS S